Agent Noise Cancellation

How the extension works. Written so you can follow it without reading the code. Version 0.2.0.

1. What this is

This is a Chrome extension for calls that run in the browser, such as Google Meet or Microsoft Teams. It does not place the call. The call page still does that. The extension cleans the audio around that call.

The toolbar window has one control: turn cleaning on or off. It also shows a status: Off, Starting, On, Error, or No call tab. There is no microphone list. The extension uses the microphone the call is already using.

2. What you hear, and what the other person hears

On. The other person receives your voice after noise removal. You hear their voice after the same kind of noise removal. The call itself stays connected.

Off. Your original microphone is put back on the call, and their audio is put back on the call page's own player. You do not have to hang up or rejoin. Noise removal stops.

3. How it works, step by step

1. You open the call.

The call page asks the browser for your microphone and connects that audio to the other person. Their audio arrives in a player on the page.

2. You open the extension and turn it on.

The popup talks to the browser tab in front of you. If that tab is not a normal web page, the popup says there is no call tab.

3. The extension enters the page.

A small script has been on the page since it loaded. Turning the extension on tells that script to start. It watches two things: the microphone the call asks for, and the player the call uses for the other person.

4. Your microphone is cleaned.

A copy of your microphone goes through two models, one after the other. The call page is then given that cleaned microphone, and the track that actually goes to the other person is swapped to it if the page had already attached the original one.

5. The other person's audio is cleaned.

Their player is routed through the same two models, and you hear the result. This incoming cleaning is on by default whenever the extension is on.

6. You turn it off.

The original microphone is put back on the call. Their player goes back to the unprocessed audio. The models are released.

4. The two models

Both run inside the browser, in real time, at 48,000 samples per second. They do not need a recording of your voice, and they do not train on your voice.

- **DeepFilterNet3** runs first. It removes background noise and the room echo that makes a voice sound distant.
- **Jitsi RNNoise** runs second. It takes another pass at leftover noise.

While the extension rewrites the microphone request, it turns the browser's own noise suppression and automatic volume off, so those do not stack on top of the models and make the voice sound over-processed. It leaves the browser's echo-cancellation setting as the call page requested it.

5. The path from the button to the call

Popup
↓ remembers On or Off, and sets the toolbar badge
Service worker
↓ carries the command into the open tab
Content script
↓ watches the microphone and the call
Page bridge
↓ DeepFilterNet3, then RNNoise
The call

- **Popup.** The small window from the toolbar. It sends On or Off and shows the status.
- **Service worker.** The extension's background. It remembers whether you turned it on and updates the toolbar badge.
- **Content script.** Runs inside the call tab and carries messages from the extension into the page.
- **Page bridge.** Sits in the call page. It sees when the page asks for a microphone, when that microphone is attached to the call, and when the page plays the other person.
- **The models.** They clean your outgoing microphone and, by default, the incoming voice you hear.

6. Echo

Echo here means a delayed copy of a sound, not ordinary background noise. The extension does not run a separate echo canceller of its own.

The noise-removal graph for your microphone does not play that cleaned audio through your speakers. A silent connection only keeps the audio engine awake. If the call page itself previews the microphone it was given, you can still hear yourself, or a knock on the table, a moment later, because that preview is now the cleaned and slightly delayed microphone.